The three-equation melt condition at an ice–water interface

and its form in `solver2d`

1 Geometry and notation

The ice occupies $y < 0$ and the water $0 < y < L_y$; the ice–water interface is the wall $y = 0$ (in `solver2d`: `bc_bot = 'noslip'`, `bc_top = 'freeslip'`, gravity along $+y$). The unit normal $\mathbf{n} = \mathbf{e}_y$ points from the interface into the water. The subscript b denotes values at the interface ($y = 0$). The melt (ablation) rate m is the speed at which the interface recedes into the ice (in $\mathrm{m\,s^{-1}}$ of ice), positive for melting. The effect of pressure on the freezing point is neglected throughout (it would add a term $\lambda_3 p_b$, with $\lambda_3 \approx -7.5 \times 10^{-4}\,^\circ\mathrm{C\,dbar^{-1}}$, to (1)).

symbol	meaning	typical value
$T,\ S$	water temperature, salinity	
$T_b,\ S_b$	temperature and salinity at the interface	unknown
m	melt rate (ice thickness per unit time)	unknown
λ_1	liquidus slope	$\approx -5.7 \times 10^{-2}\ ^\circ\mathrm{C\,psu^{-1}}$
λ_2	liquidus offset	$\approx 8.3 \times 10^{-2}\ ^\circ\mathrm{C}$
L_f	latent heat of fusion	$3.34 \times 10^5\ \mathrm{J\,kg^{-1}}$
$c_w,\ c_i$	specific heat of water, ice	$\approx 3974,\ 2009\ \mathrm{J\,kg^{-1}K^{-1}}$
$\rho_w,\ \rho_i$	density of water, ice	$\approx 1028,\ 917\ \mathrm{kg\,m^{-3}}$
$\kappa_T,\ \kappa_S$	molecular diffusivities of heat and salt in water	$\approx 1.4 \times 10^{-7},\ \approx 7 \times 10^{-10}\ \mathrm{m^2 s^{-1}}$
κ_i	thermal diffusivity of ice	$\approx 1.1 \times 10^{-6}\ \mathrm{m^2 s^{-1}}$
S_i	salinity of the ice	≈ 0

The liquidus constants differ slightly between sources; the values above are representative of those used in ice–ocean models (e.g. Holland & Jenkins 1999). Check them, and the other material properties, against the source you use.

2 The three equations

1. Freezing point (liquidus). The interface is at the local freezing temperature:

$$T_b = \lambda_1 S_b + \lambda_2. \tag{1}$$

2. Heat balance. The heat conducted from the water to the interface melts ice and is partly conducted into the ice:

$$\rho_w c_w \kappa_T \left.\frac{\partial T}{\partial y}\right|_b = \rho_i L_f\, m \ +\ \rho_i c_i \kappa_i \left.\frac{\partial T_i}{\partial y}\right|_b . \tag{2}$$

The last term (conduction into the ice, T_i the ice temperature) is often neglected, or modelled as $\rho_i c_i \kappa_i (T_b - T_s)/h_i$ with T_s the temperature at the far side of an ice layer of thickness h_i.

3. Salt balance. The ice holds (almost) no salt. The salt that the meltwater dilutes at the interface must be supplied by diffusion in the water:

$$\rho_w \kappa_S \left.\frac{\partial S}{\partial y}\right|_b = \rho_i \, m \, (S_b - S_i) \;\approx\; \rho_i \, m \, S_b. \tag{3}$$

Equations (1)–(3) determine the three unknowns T_b, S_b and m. For melting ($m > 0$) the interface is fresher than the water ($\partial S/\partial y|_b > 0$) and warmer than the freezing point of the far-field water. Since $\kappa_T/\kappa_S \approx 10^2$, salt diffusion limits the melt rate together with the heat supply.

Signs. With $\mathbf{n} = \mathbf{e}_y$ pointing into the water, the conductive fluxes *towards* the interface are $\rho_w c_w \kappa_T \, \partial T/\partial y|_b$ and $\rho_w \kappa_S \, \partial S/\partial y|_b$. This holds for the set-up above: ice at $y = 0$, water at $y > 0$, gravity along $+y$ (the ice is physically on top). If instead the ice is at $y = L_y$ (water at $y < L_y$, normal $-\mathbf{e}_y$), replace $\partial/\partial y$ by $-\partial/\partial y$.

Wall-normal velocity at the interface. As in Wells & Worster (2011), we use the reference frame in which the interface is fixed at $y = 0$ (it is quasi-planar and recedes slowly). In this frame the ice moves towards the interface at speed m, and melts there. Mass conservation across the interface, $\rho_i \, m = \rho_w \, v(0)$, gives the boundary condition

$$v(0) = \frac{\rho_i}{\rho_w} \, m \;>\; 0, \qquad u(0) = 0. \tag{4}$$

The velocity at the interface is positive: meltwater flows from the interface into the water. It is the ablation rate times ρ_i/ρ_w (≈ 0.89 for sea water, ≈ 0.92 for fresh water), i.e. it equals the volume flux of meltwater per unit area. It combines the advection of the water past the receding interface and the expansion of the ice on melting. This is condition (3.5a) of Wells & Worster (2011), $v = -(\rho_s/\rho_l)\,\dot{a}$ with $\dot{a} = -m$. (In the frame of the ice the water velocity at the interface is $v(0) - m = (\rho_i/\rho_w - 1)\, m < 0$: the volume contracts on melting.)

The interface is therefore *not* impermeable. The heat and salt balances (2) and (3) nevertheless contain only the diffusive fluxes on the water side, in either frame. In the interface-fixed frame the salt balance reads $\rho_i \, m \, S_i = \rho_w v(0) \, S_b - \rho_w \kappa_S \, \partial S/\partial y|_b$ (salt carried in by the ice = salt carried away by advection and diffusion); with (4) this is exactly (3). The same holds for heat.

3 Special cases

Stefan condition (fresh water). Without salt, (3) drops out and (1) becomes $T_b = T_m$ ($0\,^\circ\mathrm{C}$):

$$T_b = T_m, \qquad \rho_i L_f \, m = \rho_w c_w \kappa_T \left.\frac{\partial T}{\partial y}\right|_b - \rho_i c_i \kappa_i \left.\frac{\partial T_i}{\partial y}\right|_b. \tag{5}$$

Transfer-coefficient form (unresolved boundary layers). Ocean models do not resolve the diffusive sublayers and write the fluxes with transfer coefficients γ_T, γ_S and the friction velocity u_*:

$$\kappa_T \left.\frac{\partial T}{\partial y}\right|_b \to \gamma_T u_* (T_w - T_b), \qquad \kappa_S \left.\frac{\partial S}{\partial y}\right|_b \to \gamma_S u_* (S_w - S_b), \tag{6}$$

with T_w, S_w the values in the water outside the sublayer. Neglecting conduction into the ice and with $S_i = 0$, eliminating m and T_b gives a quadratic for S_b:

$$a \, S_b^2 + b \, S_b + c = 0, \qquad \begin{aligned} a &= c_w \gamma_T \lambda_1, \\ b &= -\left[c_w \gamma_T (T_w - \lambda_2) + L_f \gamma_S\right], \\ c &= L_f \gamma_S \, S_w, \end{aligned} \tag{7}$$

Since $\lambda_1 < 0$, $a < 0$ while $c > 0$, so the two roots have opposite signs and the physical root is the positive one (it satisfies $0 < S_b \le S_w$ for melting). Then T_b follows from (1) and $m = (\rho_w/\rho_i)\,\gamma_S u_*(S_w - S_b)/S_b$.

In a resolved simulation the gradients are computed directly at the wall and no transfer coefficients are needed. If the wall gradient is linearised as $\partial T/\partial y|_b \approx (T_1 - T_b)/\delta_T$ (and similarly for S), the same quadratic (7) follows with $\gamma_T u_* \to \kappa_T/\delta_T$, $\gamma_S u_* \to \kappa_S/\delta_S$, $T_w \to T_1$ and $S_w \to S_1$.

4 Non-dimensional form in `solver2d`

`solver2d` uses a length unit ℓ, a velocity unit U, and $Re = U\ell/\nu$. The diffusivities are $\kappa_T = U\ell/(Re\,Pr)$ and $\kappa_S = U\ell/(Re\,Sc)$. Write the scalars as

$$T = T_{\mathrm{ref}} + \Delta T\,\theta, \qquad S = S_{\mathrm{ref}} + \Delta S\,s, \tag{8}$$

(θ and s are scalars 1 and 2 of the code) and the melt rate as a water velocity in units of U:

$$m^* = \frac{\rho_i\, m}{\rho_w\, U}. \tag{9}$$

With y in units of ℓ, equations (1)–(3) become

$$\theta_b = \Lambda\, s_b + \theta_f, \qquad \Lambda = \frac{\lambda_1 \Delta S}{\Delta T}, \quad \theta_f = \frac{\lambda_1 S_{\mathrm{ref}} + \lambda_2 - T_{\mathrm{ref}}}{\Delta T}, \tag{10}$$

$$m^* = St\left(\frac{1}{Re\,Pr}\,\frac{\partial\theta}{\partial y}\bigg|_b - q_i\right), \qquad St = \frac{c_w \Delta T}{L_f}, \tag{11}$$

$$\frac{1}{Re\,Sc}\,\frac{\partial s}{\partial y}\bigg|_b = m^*\,(s_b + \sigma), \qquad \sigma = \frac{S_{\mathrm{ref}} - S_i}{\Delta S}, \tag{12}$$

with the Stefan number St and the non-dimensional conduction into the ice

$$q_i = \frac{\rho_i c_i \kappa_i}{\rho_w c_w U \Delta T}\,\frac{\partial T_i}{\partial y}\bigg|_b \qquad (q_i = 0 \text{ if conduction into the ice is neglected}). \tag{13}$$

Equation (12) uses $S_b - S_i = \Delta S\,(s_b + \sigma)$. If S_{ref} is chosen as the ice salinity ($S_{\mathrm{ref}} = S_i$, e.g. 0), then $\sigma = 0$.

The non-dimensional melt rate is exactly the non-dimensional wall-normal velocity at the interface, (4):

$$\frac{v(0)}{U} = m^* > 0. \tag{14}$$

The new non-dimensional parameters are St, Λ, θ_f, σ (and q_i). In `solver2d` the buoyancy enters through the Richardson numbers `ri(1)`, `ri(2)` of θ and s, and the diffusivities through `pr(1)` $= Pr$ and `pr(2)` $= Sc$.

5 Use as a boundary condition (not yet implemented)

The conditions hold at every point x of the interface and are nonlinear ($m^* s_b$) and coupled (θ and s). A practical implementation in `solver2d`:

1. At each Runge–Kutta substep, evaluate $\partial\theta/\partial y|_b$ and $\partial s/\partial y|_b$ in physical space (all x), from the current fields.

2. At each x, solve (10)–(12) for θ_b, s_b, m^*, e.g. with the linearised wall gradients of Sec. 3, which gives the quadratic (7) at every point.

3. Transform $\theta_b(x)$ and $s_b(x)$ to Fourier space and impose them as Dirichlet values on all Fourier modes in the (batched) scalar solve. The code currently imposes wall values on the mean mode only, so this needs per-mode wall values.

4. Lag the condition by one substep, or iterate it within the substep; the interface values change slowly compared with the flow.

5. Impose the wall-normal velocity $v(0,x) = m^*(x)$, $u(0,x) = 0$ (instead of $u = v = 0$). In the velocity–vorticity solver this is an inhomogeneous Dirichlet value of v at the wall, handled by the same superposition as the other wall conditions.

6. Mass conservation: in 2D the mean ($k_x = 0$) part of continuity gives $\partial\bar{v}/\partial y = 0$, so the mean blowing velocity $\bar{v} = \overline{m^*}$ is the same at every y. The wall $y = L_y$ must then let this flux out ($\bar{v}(L_y) = \overline{m^*}$, an outflow) instead of being impermeable; at present `solver2d` has $\bar{v} \equiv 0$. Equivalently, the whole water column moves away from the ice at $\overline{m^*}$, as the far field does in the frame of the interface.

7. Output the melt rate $m^*(x,t)$ and its average.

6 Comparison with Wells & Worster (2011)

Wells & Worster (2011) write the interface as $y = a(t)$, with the solid at $y < a$ and the liquid at $y > a$, so that $\dot{a} = \partial a/\partial t < 0$ for ablation; subscripts s (solid) and l (liquid), solute concentration C and solutal diffusivity D. With pressure neglected, their interface conditions are identical to (1)–(4) under the correspondence

$$\dot{a} = -m, \qquad \rho_s = \rho_i, \qquad \rho_l = \rho_w, \qquad c_l = c_w, \qquad \kappa_l = \kappa_T,$$
$$D = \kappa_S, \qquad C = S, \qquad C_s = S_i, \qquad T_i^{\mathrm{WW}} = T_b, \qquad C_i = S_b.$$

	Wells & Worster (2011)		this document
liquidus	$T_i = T_m - \Gamma\,(C_i - C_s)$	(2.1)	$T_b = \lambda_1 S_b + \lambda_2$: $\lambda_1 = -\Gamma$, $\lambda_2 = T_m + \Gamma C_s$
heat	$\rho_s L\dot{a} = \rho_s c_s \kappa_s T_y\|_{a^-} - \rho_l c_l \kappa_l T_y\|_{a^+}$	(2.2)	(2) with $\dot{a} = -m$
salt	$\rho_s (C_i - C_s)\,\dot{a} = -\rho_l D\, C_y\|_{a^+}$	(2.3)	(3) with $\dot{a} = -m$
velocity	$v(0) = -(\rho_s/\rho_l)\,\dot{a}$	(3.5a)	(4): $v(0) = (\rho_i/\rho_w)\,m$

Remarks:

- Liquidus: Wells & Worster use $T_m = 0\,^\circ\mathrm{C}$ and $\Gamma \approx 0.06\,^\circ\mathrm{C\,psu^{-1}}$ with $C_s \approx 0$, i.e. $\lambda_1 \approx -0.06\,^\circ\mathrm{C\,psu^{-1}}$ and $\lambda_2 = 0$. Holland & Jenkins (1999) fit $\lambda_1 \approx -0.057\,^\circ\mathrm{C\,psu^{-1}}$ and $\lambda_2 \approx 0.083\,^\circ\mathrm{C}$ over the oceanic salinity range; the difference in T_b is below $0.1\,^\circ\mathrm{C}$.

- Heat: Wells & Worster show (their appendix) that the conductive heat flux into the solid is typically small and neglect it, i.e. $q_i = 0$ in (11).

- Velocity: in the interface-fixed frame the wall-normal velocity at $y = 0$ equals the meltwater volume flux $(\rho_i/\rho_w)\,m$ and is positive (directed into the water). It equals the ablation rate m itself only if $\rho_i = \rho_w$.

- Their ablation rate, (3.15), $\dot{a} = -(\rho_l/\rho_s)\,v(0)$, is (4) solved for m. They also note that the fluid flow, and $v(0)$ itself, do not depend on ρ_s/ρ_l; only the conversion from $v(0)$ to the ablation rate does. This agrees with (11), which determines $m^* = v(0)/U$ without ρ_i/ρ_w.

- Buoyancy: Wells & Worster keep only the compositional (salt) buoyancy, $\rho = \rho_\infty + \rho_l\beta(C - C_\infty)$; `solver2d` has both, through `ri(1)` and `ri(2)` (set `ri(1)` $= 0$ to reproduce their assumption).

References

- H. H. Hellmer and D. J. Olbers, A two-dimensional model for the thermohaline circulation under an ice shelf, *Antarctic Science* 1 (1989) 325–336.
- D. M. Holland and A. Jenkins, Modeling thermodynamic ice–ocean interactions at the base of an ice shelf, *J. Phys. Oceanogr.* 29 (1999) 1787–1800.
- A. J. Wells and M. G. Worster, Melting and dissolving of a vertical solid surface with laminar compositional convection, *J. Fluid Mech.* 687 (2011) 118–140.